PingSheng Kao

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SUMMARY

ML Data Scientist with 5+ years of experience building evaluation systems for multimodal, LLM, and agentic products. Designs expert-aligned LLM judges, factual-consistency and tool-call evaluations, ground-truth workflows, user simulations, and scalable batch evaluation processes that turn model failures into actionable product-quality metrics.

EXPERIENCE

ML Data Scientist - **Apple** July 2020 - Present
Santa Clara, California

- Built and scaled **LLM-as-judge and agent-as-judge** workflows on internal batch infrastructure, auditing long multimodal conversations for **factual consistency and tool-call behavior**, debugging throughput and rate-limit issues, and surfacing evaluator results in dashboards.
- Built **expert-aligned evaluation** for a multimodal health-related AI product: an auditable, reference-based multi-stage LLM evaluator, gold-set curation and labeling workflows, and DSPy-style iteration that reached **90% alignment with experts on a 40-example gold set**.
- Built a **simulated-user agent**, targeted scenario generators, and user-centered metrics to stress-test a multimodal conversational agent, probing visual-memory and sycophancy failures and estimating rates of user-intent misalignment and false inferred-state updates at scale.
- Wrote **SQL** in an evaluation pipeline querying **AWS Athena** to calculate model-quality metrics and export results to dashboards; combined those outputs with release-level failure analysis to attribute regressions and produce decision-ready findings for technical leadership.
- Developed **scene-graph-based evaluation** for free-form image descriptions, enabling fine-grained measurement of OCR, counting, attribute, and spatial-reasoning failures in multimodal-model outputs.
- Developed embedding-driven failure discovery for **Vision Pro hand tracking** using similarity search, clustering, OOD, and position-level analysis; contributed to a quality scorecard with final missing-pinch rate below 2% and false positives below 5 per hour.

RESEARCH CONTRIBUTION

GH-ESD: Grounded Hypothesis-Driven Error Slice Discovery for Instance-Level Vision Tasks 2025
Experiment contributor (credited as Sam Kao) - accepted to ECCV - arXiv:2512.24592

- Contributed experiments to research using LLMs and VLMs to generate and verify grounded failure hypotheses for fine-grained, interpretable error-slice discovery.
- Trained and fine-tuned vision models on VLM-discovered failure slices and benchmarked targeted repair against alternative slice-discovery approaches.

ADDITIONAL EXPERIENCE

Algorithm and Architecture Intern - Deep Learning Compression - **Ambarella** June 2019 - August 2019
Santa Clara, California

- Developed a saliency-based neural-network quantization heuristic that reduced an exponential search formulation to a linear-time method, and proposed a mixed-precision algorithm integrated into company software.

EDUCATION

M.S. in Computer Science and Engineering - **University of Michigan, Ann Arbor** 2018 - 2020
B.S. in Electronic Engineering - **National Taiwan University** 2013 - 2017
GPA: 3.9/4.0

SKILLS

Programming	Python, SQL, C++, JavaScript
Evaluation	LLM-as-Judge, Agent-as-Judge, Ground-Truth Dataset Design, Factual-Consistency Evaluation, Tool-Call Evaluation, Human Evaluation
Data & ML	AWS Athena, PyTorch, DSPy, Statistical Analysis, Experiment Design, Embeddings, Clustering, Tableau
Multimodal & Agents	VLM/MLLM Evaluation, User Simulation, Failure Analysis, Scene Graphs, AI-Assisted Development with Claude Code